

## Lab 1: Requirements Engineering & UML Use-Case Modelling

*Problem Statement #18 — Healthcare & Telemedicine*

### Clinical Trial Subject Recruitment Engine

Name: Aditya Singhal

SRN: PES1UG24AM420

Section / Batch: AIML-H

#### 1. Problem Context & Overview

Medical research institutions require an eligibility pre-screening platform for clinical trials that tracks subject inclusion/exclusion criteria, informed-consent milestones, and visit compliance logs.

Target Stakeholders / Actors: Trial Candidate, Clinical Researcher, and Principal Investigator (added to satisfy the minimum of three actors required for the use-case model; the Principal Investigator reviews audit and compliance reports).

#### Deliverables in this Document

- Complete Requirements Table — 5 Functional Requirements (FR-001–FR-005) and 2 Non-Functional Requirements (NFR-001–NFR-002).
- UML Use-Case Diagram — all actors, primary use cases, one «include» relationship, and one «extend» relationship.
- Use-Case Flow Specification — Preconditions, Postconditions, Main Success Scenario, and one Alternate Flow for the core use case “Complete Eligibility Screening Survey.”

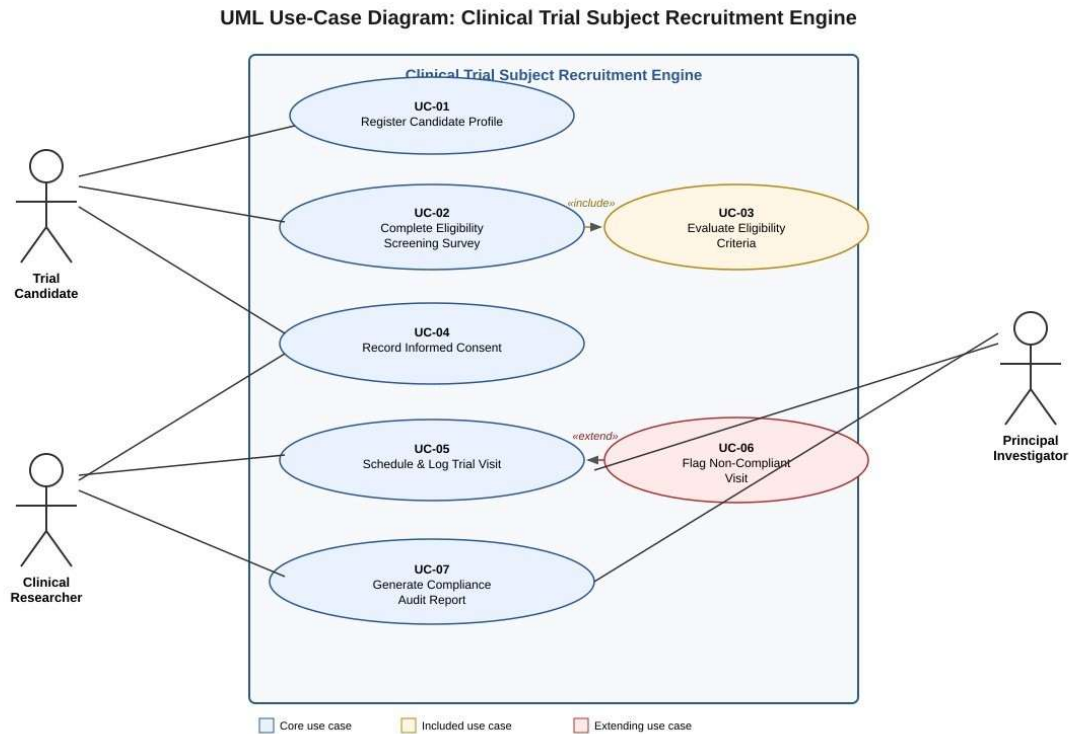
## 2. Requirements Table

FR-001 and NFR-001 are the requirements provided in the problem statement; FR-002–FR-005 and NFR-002 were elicited to complete the specification.

| Req ID  | Type                                   | Description                                                                                                                                                                                         | Priority | Acceptance Criteria                                                                                                                                                                                  | Rationale                                                                                                                                                 |
|---------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-001  | Functional                             | The system shall evaluate candidate survey responses against trial protocol inclusion/exclusion criteria and classify eligibility status.                                                           | High     | Pass: Ineligible candidate flagged with specific failed criteria. Fail: Subject with exclusion criteria marked eligible.                                                                             | Core eligibility screening function; prevents ineligible or high-risk candidates from being enrolled, protecting subject safety and trial data integrity. |
| FR-002  | Functional                             | The system shall allow a Trial Candidate to create and submit a registration profile including demographic data, medical history, and contact information.                                          | High     | Pass: Profile saved with a unique Candidate ID and all mandatory fields validated before submission. Fail: Profile submitted with missing fields or a duplicate Candidate ID.                        | Establishes the auditable baseline record required before any screening or consent activity can occur.                                                    |
| FR-003  | Functional                             | The system shall capture and timestamp each informed consent milestone (document viewed, e-signature applied, witness confirmation) for every candidate.                                            | High     | Pass: Consent record shows all required milestones with timestamps and signer identity before status changes to “Consented.” Fail: Candidate reaches “Consented” status with a missing milestone.    | Documented, timestamped consent is a regulatory requirement (ICH-GCP) and forms part of the trial's legal record.                                         |
| FR-004  | Functional                             | The system shall allow a Clinical Researcher to schedule trial visits and log each visit's actual date, attendance status, and protocol deviations.                                                 | Medium   | Pass: A missed or late visit (>48 hours from scheduled date) is automatically flagged in the compliance log. Fail: A missed visit is not flagged within 48 hours.                                    | Visit compliance tracking detects protocol deviations early and preserves data quality for trial outcomes.                                                |
| FR-005  | Functional                             | The system shall generate an audit-trail report, on demand, summarizing eligibility decisions, consent status, and visit compliance for a candidate or site.                                        | Medium   | Pass: Report generates in under 10 seconds and includes a cryptographic hash for tamper-evidence. Fail: Report omits any eligibility, consent, or compliance record for the selected candidate/site. | Supports regulatory inspection readiness and gives Principal Investigators a consolidated compliance view.                                                |
| NFR-001 | Nonfunctional (Performance & Security) | The recruitment database must comply with 21 CFR Part 11 electronic record standards, including cryptographic audit trails.                                                                         | High     | Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.                                                                                                    | Regulatory mandate for electronic records in FDA-regulated clinical trials; non-compliance risks trial data being ruled inadmissible.                     |
| NFR-002 | Nonfunctional (Security & Privacy)     | The system shall enforce role-based access control (RBAC) so only authenticated Clinical Researchers and Principal Investigators can view or export candidate PII, with all access attempts logged. | High     | Pass: An unauthorized access attempt is denied and logged (user ID, timestamp, resource) within 1 second. Fail: A user without an assigned role successfully views PII.                              | Protects candidate privacy under HIPAA/dataprotection regulations and limits the audit surface for compliance reviews.                                    |

### 3. UML Use-Case Diagram

Actors: Trial Candidate, Clinical Researcher, Principal Investigator. The diagram models seven use cases (UC01–UC-07), including one «include» relationship (UC-02 includes UC-03) and one «extend» relationship (UC06 extends UC-05).



### 4. Use-Case Flow Specification

#### Use Case: UC-02 — Complete Eligibility Screening Survey

##### Actors

Trial Candidate (primary); Clinical Researcher (notified on completion).

##### Preconditions

- The candidate has a registered profile (UC-01 completed) with a valid Candidate ID.
- The trial protocol's inclusion/exclusion criteria are configured in the system.
- The candidate is authenticated and logged into the self-service portal.

##### Postconditions

- Success: The candidate's eligibility status is recorded as “Eligible” or “Ineligible,” together with the specific criteria evaluated, and the decision is timestamped in the audit trail.

- Failure: No status change is persisted; the candidate remains “Pending Screening.”

### **Main Success Scenario**

1. Candidate selects “Start Eligibility Screening” from the portal dashboard.
2. System retrieves the active trial protocol's inclusion/exclusion criteria and displays the screening survey.
3. Candidate answers each survey question (medical history, current medications, demographic criteria, etc.).
4. Candidate submits the completed survey.
5. System validates that all mandatory questions have been answered.
6. System invokes UC-03 “Evaluate Eligibility Criteria” («include») to compare responses against protocol criteria.
7. System classifies the candidate as “Eligible” or “Ineligible” and records the specific criteria matched or failed.
8. System timestamps the eligibility decision and stores it in the candidate's audit trail.
9. System displays the eligibility outcome to the candidate and notifies the assigned Clinical Researcher.
10. Use case ends successfully.

### **Alternate Flow — 5a. Incomplete Survey Submission**

- 5a1. System detects one or more mandatory questions left unanswered.
- 5a2. System highlights the missing fields and displays a validation message.
- 5a3. Candidate completes the missing fields and resubmits (flow returns to step 5).
- 5a4. If the candidate abandons the survey without resubmitting, the system saves a draft and the use case ends without a status change.